

PDBsum1 on the Lab VM

Quick guide

What it does

Give PDBsum1 a PDB file (crystal structure, docking result or model) and it builds a set of linked web pages summarising it: the chains and their secondary structure, **ligand pages with a LIGPLOT** of every hydrogen bond and hydrophobic contact, **protein–protein interfaces**, surface **clefts** (possible pockets) and quality checks (Ramachandran plot). A 2-chain structure with a ligand takes about a minute.

How to use it

You need SSH access to the lab VM (ask Dr. Bilal for the address and login).

1. Log in and go to the tool

```
ssh <your-login>@<lab-vm-address>  
cd /mnt/bilal/pdbsum1
```

2. Get your PDB file onto the VM — from your own computer, in a second terminal:

```
scp myprotein.pdb <your-login>@<lab-vm-address>:/mnt/bilal/pdbsum1/input/
```

3. Run it (on the VM)

```
./pdbsum1.sh myprotein.pdb
```

Progress lines (“Running ...”) scroll by; it ends with “Results at ...”.

4. Copy the results to your computer (from your own computer)

```
scp -r <your-login>@<lab-vm-address>:/mnt/bilal/pdbsum1/results ./pdbsum1-results
```

Open **pdbsum1-results/index.html** in any browser. No software is needed to view it, and you can zip the folder and send it to colleagues.

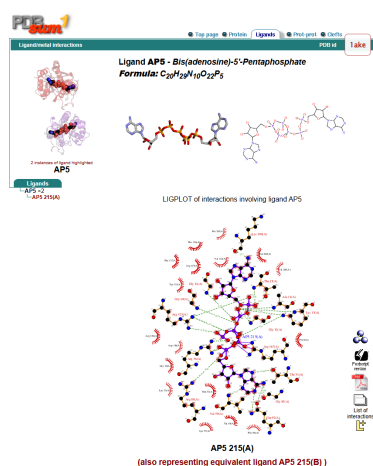
Good to know

- **Give your run a name.** Add a 4-character code after the file: `./pdbsum1.sh model17.pdb m7a1`. A file named exactly four characters (**1ake.pdb**) uses that name; otherwise runs are numbered **a001**, **a002**, ...

- **Many structures:** put all the files in `input/`, write a text file `input/list.txt` with one line per structure (`file.pdb code`), then run `./pdbsum1.sh -l list.txt`. All appear on the front page.
- **Long runs:** start `tmux new -s pdbsum1` before running, so it keeps going if your connection drops (detach: `Ctrl-b` then `d`; return: `tmux attach -t pdbsum1`).
- **Better results:** add a `TITLE` line (shown on the page) and a `DBREF` line with the UniProt accession (adds the Pfam domain diagram) to your PDB file. Docking and modelling programs sometimes write non-standard PDB files; if a run fails, check the format first.
- **Big assemblies:** clefts are skipped for structures with more than 4 protein chains. Adding `-nomaxchn` forces them, but it can take very long.
- **Shared machine:** results accumulate in `results/`. Copy what you need and tell Dr. Bilal before deleting anything.

Reading the results

Click your structure on the front page. The tabs across the top are: **Top page** (overview), **Protein** (secondary structure, Ramachandran, motifs), **Ligands** (one page per ligand/metal, with the LIGPLOT), **Prot-prot** (interfaces) and **Clefts** (surface pockets). In a LIGPLOT, green dashed lines are hydrogen bonds (with distances) and red spokes are hydrophobic contacts; PostScript and PDF versions are provided for papers.



A Ligands page (PDB 1ake, ligand AP5).

If something goes wrong

A working run keeps printing progress; if nothing changes for several minutes, press `Ctrl-C` and check your input file. If you get a “permission denied” from Docker, your account has not been given Docker access yet — ask Dr. Bilal. For a missing or empty page, look in that structure’s `logs/` folder inside `results/`. For anything else, send the input file and the `logs/` folder to Dr. Bilal.

PDBsum1 is by R. A. Laskowski and colleagues (EMBL-EBI). If you use PDBsum in a publication, cite: Laskowski RA et al., *PDBsum: Structural summaries of PDB entries*, Protein Science 2018;27:129–134.